

IS 525: Final Project

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Topic: Store Sales Forecasting

Abstract:

Using interactive Tableau dashboards, this project examines past sales and profitability performance to identify patterns, regional differences, and category-level factors that influence business outcomes. In order to facilitate data-driven decision-making, the analysis focuses on comprehending sales growth patterns, profit margin behavior, and regional performance variations using transactional retail data. Three dashboards—Business Performance Overview and Product & Profitability Analysis, Forecasting Dashboard. The final dashboard extends the analysis into the future by leveraging an XGBoost predictive model to forecast weekly sales for the next ten weeks, offering visibility into expected demand patterns, regional exposure, and forecast reliability.

Together, these dashboards form a cohesive decision-support system that enables stakeholders to assess past performance, understand current profit dynamics, and proactively plan for future demand, aligning analytical insights with strategic business planning.

When combined, these dashboards provide non-technical stakeholders with actionable insights from complex data.

1. Problem Statement

Businesses frequently monitor sales performance without fully comprehending the long-term interactions between profitability, growth trends, and regional variations. Decision-makers may ignore underperforming areas or product categories until the financial impact becomes substantial, and high revenue does not always equate to high profit.

This project's main issue is how to analyze sales, profitability, and growth trends in tandem to find opportunities for improvement, risks, and strengths across time, geography, and product categories.

2. Purpose of the Project

This project's goal is to:

- Examine past sales and profit information to identify trends in performance.
- Compare trends in profitability and revenue growth.
- Determine which areas and product categories are performing well and poorly.
- Provide information via user-friendly dashboards to aid in business decision-making.

The dashboards' interactive key performance indicator exploration is intended to go beyond static reporting.

3. Intended Audience

The primary audience for this analysis includes:

- Business executives and senior management
- Sales and regional managers
- Strategy and operations teams
- Non-technical stakeholders who need high-level insights rather than raw data

The dashboards are intentionally designed to be self-explanatory, visually intuitive, and actionable for users without prior analytical or Tableau expertise.

4. Data Sources

The analysis uses a retail transactional dataset containing historical order-level data. Key fields include:

- Order date
- Sales revenue
- Profit

- Product sub-category
- Region and state

The dataset was provided for academic analysis and represents multiple years of sales activity across regions and product categories.

Source: <https://www.kaggle.com/datasets/tanayatiptre/store-sales-forecasting-dataset/data>

5. Data Preparation

The dataset was thoroughly prepared before analysis and visualization to guarantee analytical precision, consistency, and appropriateness for both descriptive and predictive modeling.

Steps for data transformation and cleaning included:

Date field standardization to allow for consistent time-series analysis on a monthly, annual, and weekly basis

Sales and profit records are validated, and any null or incomplete entries are examined and filtered as needed.

Measures at the region, subcategory, and time levels should be appropriately aggregated to support various analytical viewpoints.

To facilitate analysis, derived and calculated fields were developed:

Calculated as profit in relation to sales, profit margin

Sales growth year over year (YoY) to assess long-term performance patterns

Sales and profit moving averages are used to reduce volatility and draw attention to underlying patterns.

Preparation for predictive modeling comprised:

Weekly aggregation of past sales data for time-series forecasting

Using feature engineering, temporal patterns like seasonality and trend continuity can be captured.

Prior to forecasting, historical data is separated for model validation and training.

Because of its capacity to manage complex interactions and non-linear relationships in time-series data, an XGBoost regression model was used to predict weekly sales for the ensuing ten-week period. Forecast error analysis was used to further evaluate model outputs in order to gauge confidence across regions and subcategories.

The data was dependable, analytically sound, and appropriate for supporting both exploratory insights and forward-looking business decisions thanks to these preparation and modeling procedures.

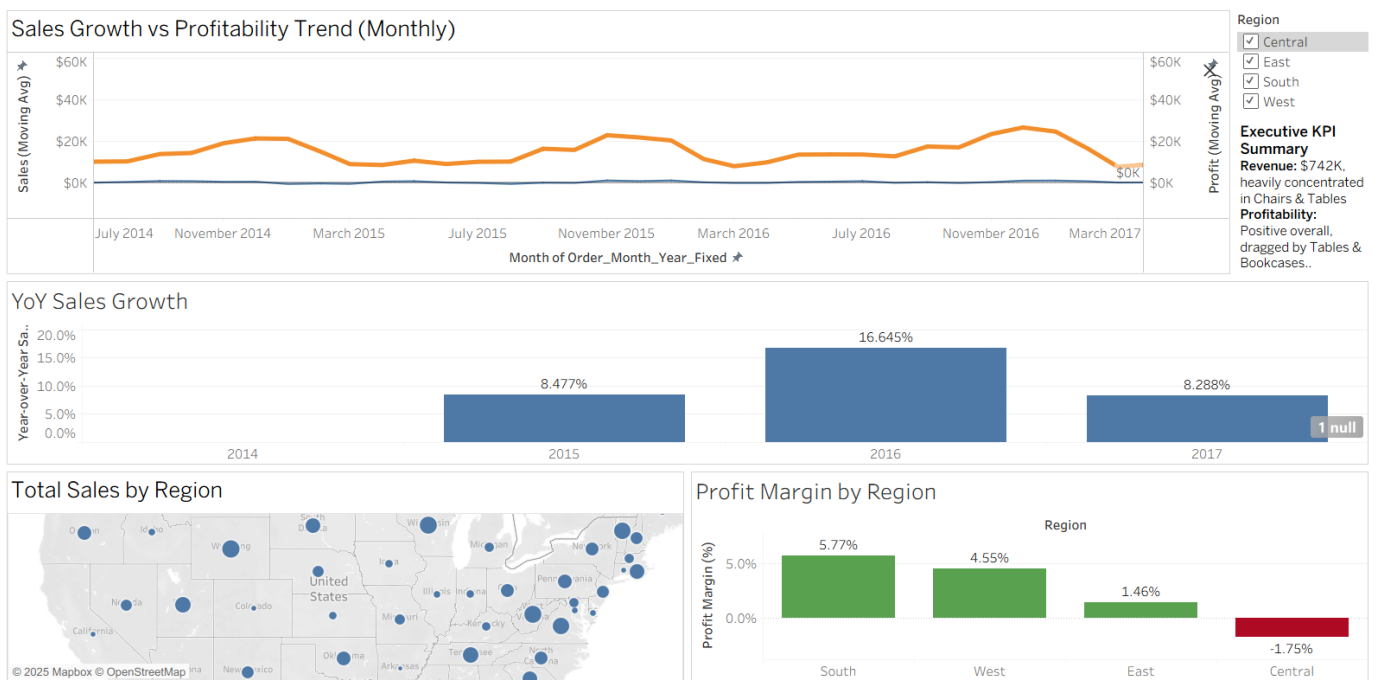
6. Analysis Methods

The project applies the following analytical techniques:

- Time-series trend analysis using monthly moving averages
- Year-over-year growth calculations to assess performance changes
- Regional comparison of sales and profit margins
- Visual segmentation using color encoding to highlight positive vs. negative performance
- Interactive filtering to allow dynamic exploration across regions and categories

All analysis was conducted using Tableau's calculated fields, table calculations, and dashboard interactivity features.

Dashboard 1: Business Performance Overview



Objective:

This dashboard combines sales growth, profitability trends, and regional performance to provide a high-level, time-based, and geographic understanding of overall business performance.

The dashboard's analytical narrative is top-down:

To find out how the company is doing overall, start with time trends.

By contrasting sales growth with profitability, you can verify the sustainability of growth.

To determine regional strengths and weaknesses, break performance down geographically.

To direct managerial attention, quantify the risks to regional profitability.

This sequence is similar to how executives usually assess performance: "Are we growing?" is the first question, followed by "Is that growth profitable?" and "Where is performance breaking down?"

Intended Stakeholders

This dashboard is primarily designed for:

- Senior leadership and executives
- Regional sales managers
- Strategy and operations teams

The visualizations are optimized for **quick interpretation**, enabling stakeholders to identify trends and risks without deep technical analysis.

Visual 1: Sales Growth vs Profitability Trend (Monthly)

Why this visual appears first: By illustrating how sales and profit change over time, this visualization lays the groundwork and provides an answer to the most important business question: How is the company doing overall?

Insight: Profit is comparatively flat and erratic, but sales show distinct seasonal fluctuations with recurrent peaks. This suggests margin pressure during times of high sales since revenue growth does not always translate into profit growth.

Visual 2: Year-over-Year (YoY) Sales Growth

Why this visual follows: This graphic zooms out to determine whether growth is increasing or decreasing annually after tracking monthly trends.

Insight: YoY sales growth slowed in 2017 after reaching a peak of roughly 16.6% in 2016. This implies that the company might be on the verge of reaching a growth plateau, indicating the need for fresh growth catalysts.

Visual 3: Total Sales by Region (Map)

Why this visual appears next: The next logical question to ask after understanding overall growth is where that revenue is coming from geographically.

Insight: There is an unequal distribution of sales among the regions, with some regions making a disproportionate contribution to overall revenue. This draws attention to the potential for geographic expansion as well as the risk associated with regional concentration.

Visual 4: Profit Margin by Region

Why this visual concludes the dashboard: Evaluating whether high-revenue areas are also profitable is the last stage.

Insight: The Central region exhibits a negative profit margin, suggesting structural inefficiencies or pricing problems despite revenue contribution, while the South and West regions maintain healthy profit margins.

Dashboard-Level Business Insights

When viewed holistically, the dashboard reveals that:

- Revenue growth exists but is not uniformly profitable
- Growth momentum peaked and has since moderated
- Regional performance varies significantly
- Some regions generate revenue at the expense of profitability

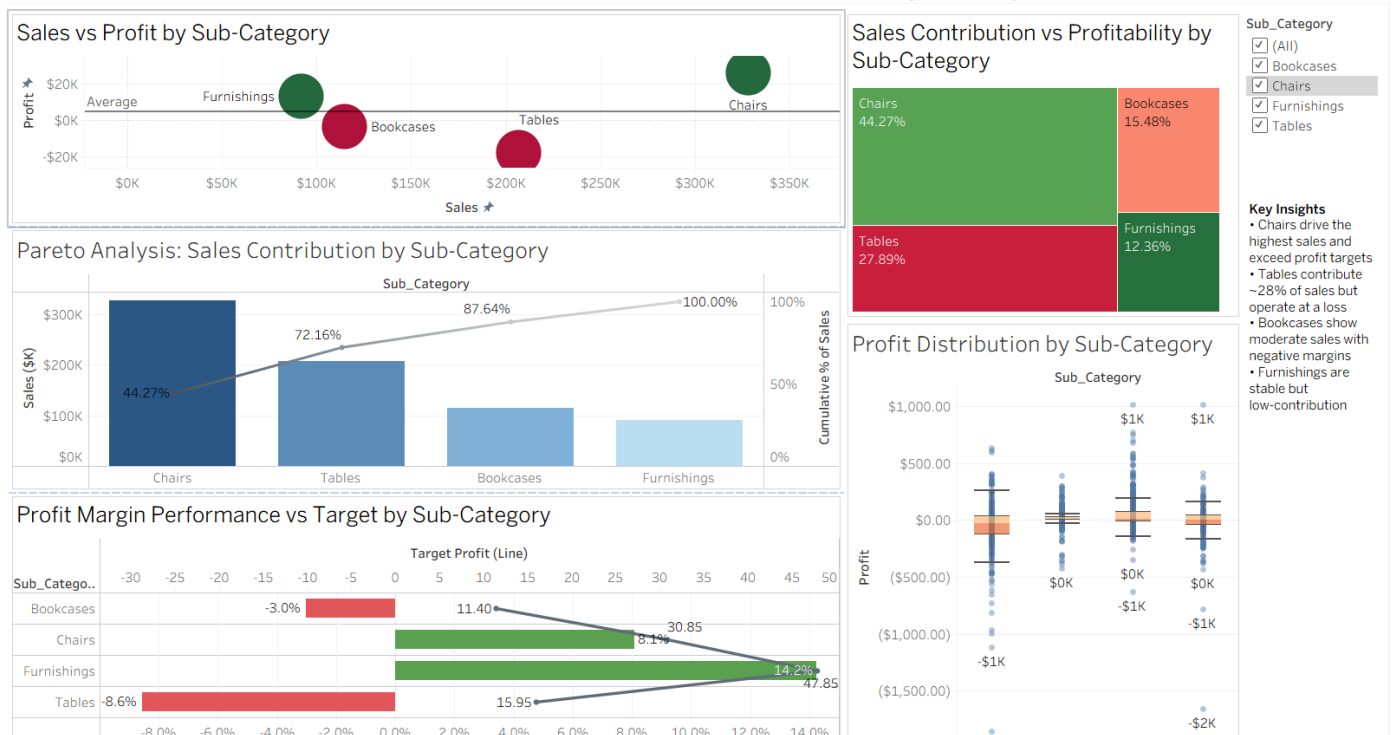
This dashboard enables leadership to move from growth monitoring to risk identification and strategic prioritization.

Transition to Dashboard 2

While Dashboard 1 explains when and where performance issues occur, it does not explain what is driving them. To identify the root causes behind profitability challenges, we next shift focus from regions and time to product-level performance.

This leads into Dashboard 2: Product & Profitability Analysis, which examines how individual product sub-categories contribute to sales and profit outcomes

Dashboard 2: Product & Profitability Analysis



Objective

This dashboard explains **what is driving profitability (or losses) at the product sub-category level**, so decision-makers can prioritize which product areas to protect, fix, or rethink.

The storyline is intentionally ordered like a manager's thought process:

1. Start by identifying which sub-categories dominate revenue (Sales contribution + profitability at a glance).
2. Then validate the revenue concentration pattern using Pareto (how dependent we are on top categories).
3. Next, compare total sales vs total profit to reveal categories that sell a lot but lose money.
4. Then examine profit volatility and risk using distribution (how consistent the profits are).
5. Finally, benchmark profitability against a target (which categories meet the expected margin standard).

Intended Stakeholders

This dashboard is most relevant for:

- Product/ category managers
- Pricing and Promotion teams
- Finance and profitability analysts
- Merchandising and operations leadership

Visual 1: Treemap of Sales Contribution vs. Profitability by Sub-Category

Why this is prioritized: It provides the quickest "portfolio snapshot," including the source of the funds and their overall health.

What it communicates:

Size = Sales contribution (who determines revenue share)

Color = Profit margin (red represents margin risk, green represents profitability)

Insights: This helps prioritize action by instantly highlighting categories that are high-revenue + low-margin (risk) versus high-revenue + healthy margin (core strengths).

Visual 2: Sales Contribution by Sub-Category Using Pareto Analysis

Why this comes in second: Pareto measures concentration to verify dependence on a few categories after observing

contribution visually.

Business insight: The majority of sales are accounted for by a small number of subcategories, indicating a clear concentration risk and the areas where improvements will have the greatest business impact.

Visual 3: Sales versus Profit by Sub-Category (Bubble/Scatter)

Why is this the third? After determining "who sells the most," we determine whether those categories are truly profitable.

Business insight: This clearly identifies "problem children", subcategories with high sales but negative profit, which is a clear indication of problems with pricing, discounts, shipping, or product costs.

Visual 4: Box Plot of Profit Distribution by Sub-Category

The scatter displays totals but lacks stability, which is why it ranks fourth. This view assesses risk and consistency.

Business insight: Wide spreads and outliers in some categories indicate inconsistent profits, which are frequently an indication of erratic discounting, returns, or shipping variability.

Visual 5: Target Comparison of Profit Margin Performance by Sub-Category

Why this comes last: Following problem diagnosis, we compare the results to the business standard to determine who achieves the desired margin and who doesn't.

Business insight: This transforms analysis into a view that is ready for decision-making, showing which subcategories are fulfilling expectations and which call for intervention (discount control, cost reduction, assortment adjustments, or price adjustments).

Dashboard-Level Summary Insight

Overall, Dashboard 2 shows that:

- Revenue is concentrated in a few sub-categories
- High sales $\neq$ high profitability
- Some categories are structurally margin-negative
- Profit performance varies in stability and target attainment

This makes the dashboard actionable for product strategy, pricing, and margin improvement.

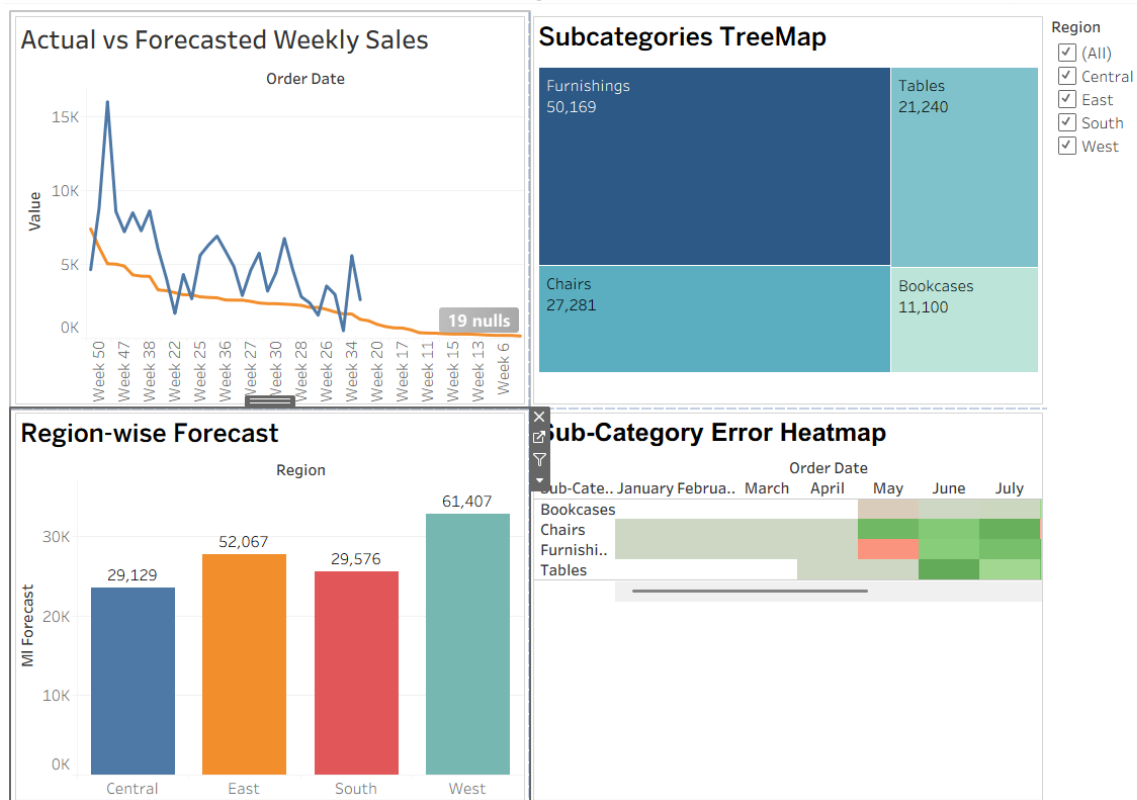
Transition to Dashboard 3

Dashboard 2 explains what categories drive profit and which categories create risk today.

The next question is forward-looking: What will future sales look like, and how can we plan inventory, staffing, and strategy accordingly?

That leads into **Dashboard 3: Sales Forecasting**.

Dashboard 3: Sales Forecasting & Future Demand Outlook (XGBoost)



Objective

After understanding historical performance (Dashboard 1) and current product-level profitability drivers (Dashboard 2), the final business question is forward-looking:

What will sales look like in the near future, and where should the business prepare for growth or risk?

This dashboard uses XGBoost, a powerful gradient boosting model, to forecast weekly sales for the next 10 weeks, leveraging historical sales patterns.

The storyline is designed to move from forecast validation → demand concentration → geographic exposure → model reliability, ensuring stakeholders trust both the predictions and the decisions derived from them.

Intended Stakeholders

- Demand planning & supply chain teams
- Sales & operations leadership
- Finance & budgeting teams
- Regional managers
- Data & forecasting analysts

Visual 1: Weekly Sales Predicted vs. Actual (XGBoost)

Why this graphic comes first: Stakeholders need to know how well the model captures past trends before they can trust any forecast.

What it communicates:

Weekly sales history compared to XGBoost forecast

The trend will continue for the next ten weeks.

The model's handling of seasonality and smoothness

Business insight: The model successfully captures demand patterns, as evidenced by the close alignment between actual and predicted values, making the short-term forecast trustworthy for making planning decisions.

Visual 2: Predicted Sales Contribution by Subcategory (Treemap)

Why this is ranked second:

Which product categories will generate future revenue after the overall forecast has been verified?

What it communicates:

Each subcategory's relative contribution to projected sales

Demand concentration across products in the future

Business insight: This ensures sufficient stock levels and targeted promotional strategies for future demand by assisting product and inventory teams in prioritizing high-impact subcategories.

Visual 3: Sales Forecast by Region

Why this is ranked third: Knowing where demand will manifest is the next step after determining what will sell.

What it communicates:

Distribution of projected sales by region

Over the course of the forecast period, regional demand intensity

Business insight: Regional managers can proactively allocate inventory, staffing, and logistics capacity to regions expected to experience higher sales growth, reducing fulfillment bottlenecks.

Visual 4: Sub-Category Forecast Error Heatmap

Why this is last:

This visual addresses a critical analytical question:

Where is the model more or less confident?

What it conveys:

- Forecast error magnitude by sub-category
- Identification of stable vs volatile demand segments

Business insight:

Categories with higher forecast error indicate greater uncertainty, signaling the need for cautious planning, safety stock buffers, or closer monitoring.

Dashboard-Level Summary Insight

Dashboard 3 transforms historical insights into **actionable foresight** by showing that:

- Short-term sales trends can be reliably forecasted using XGBoost
- Future revenue remains concentrated in a few key sub-categories
- Regional demand patterns are uneven and require tailored planning
- Forecast confidence varies across product categories

Together, this enables **data-driven demand planning and risk-aware decision-making**.

Overall Project Storyline Closure

- **Dashboard 1** established *how the business has performed historically*
- **Dashboard 2** explained *where profits are created or lost today*
- **Dashboard 3** projects *what is likely to happen next and how to prepare*

This end-to-end flow moves from **descriptive** → **diagnostic** → **predictive analytics**, aligning directly with modern business intelligence and analytics practices.